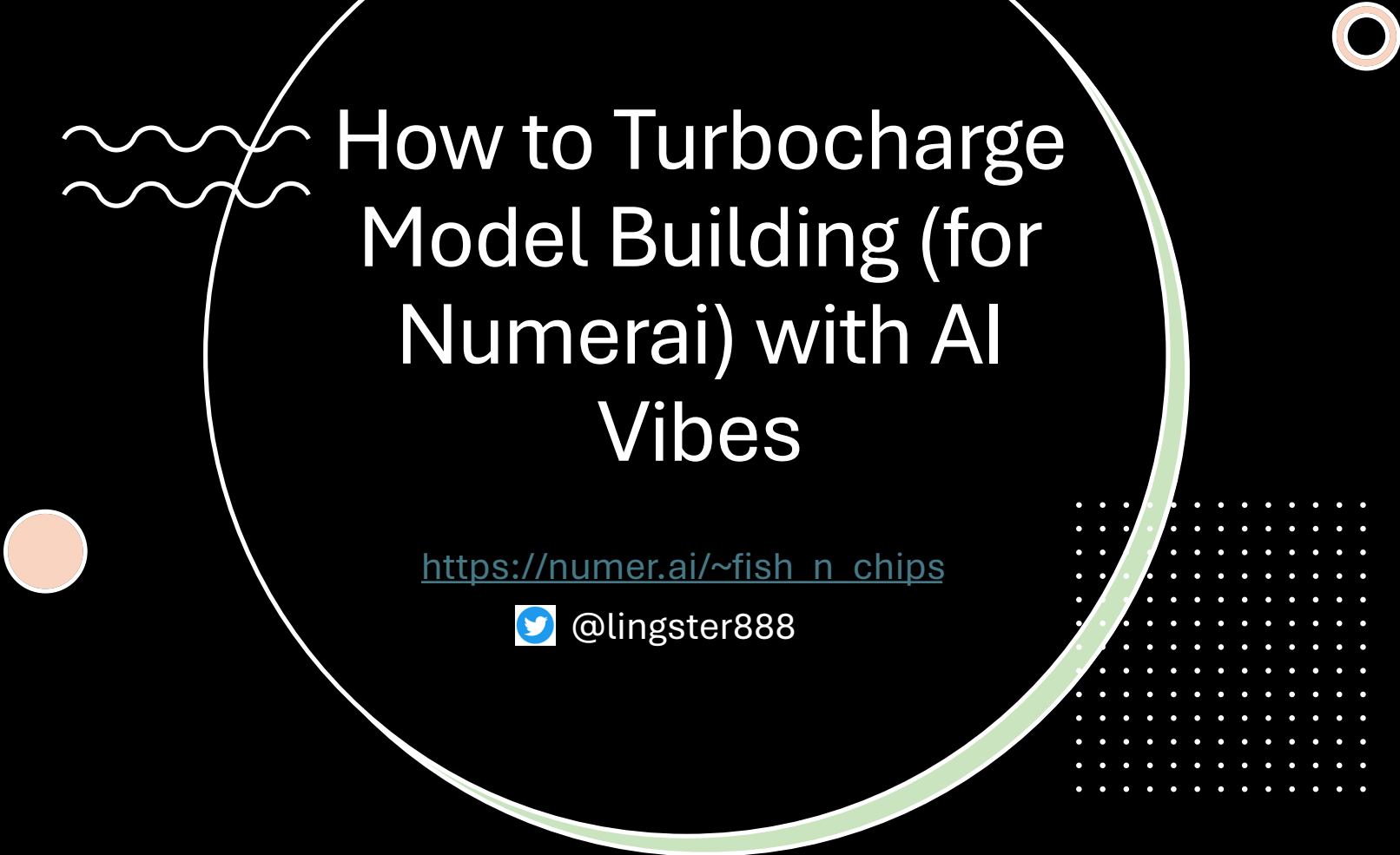




How to Turbocharge Model Building (for Numerai) with AI Vibes

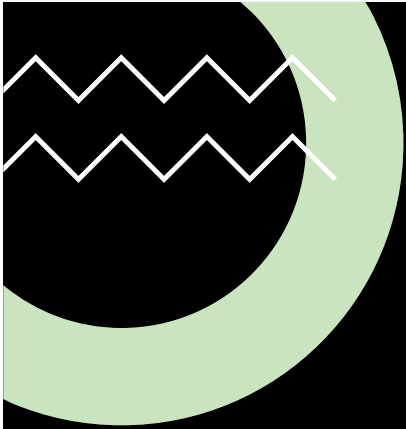
https://numer.ai/~fish_n_chips



@lingster888

History and how I got here

- Building Holly – async web based web developer
 - Am looking for help and contributors
- But could this be used to solve data science tasks?



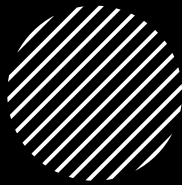
Kaggle

- Initial trial to solve the Kaggle House Price intro competition
- First attempt (zero-shot) reached 350/20k+ entrants
- Prompt engineering -> 25th position
- But is AI just regurgitating known solutions online?
- Try my hand on Numerai

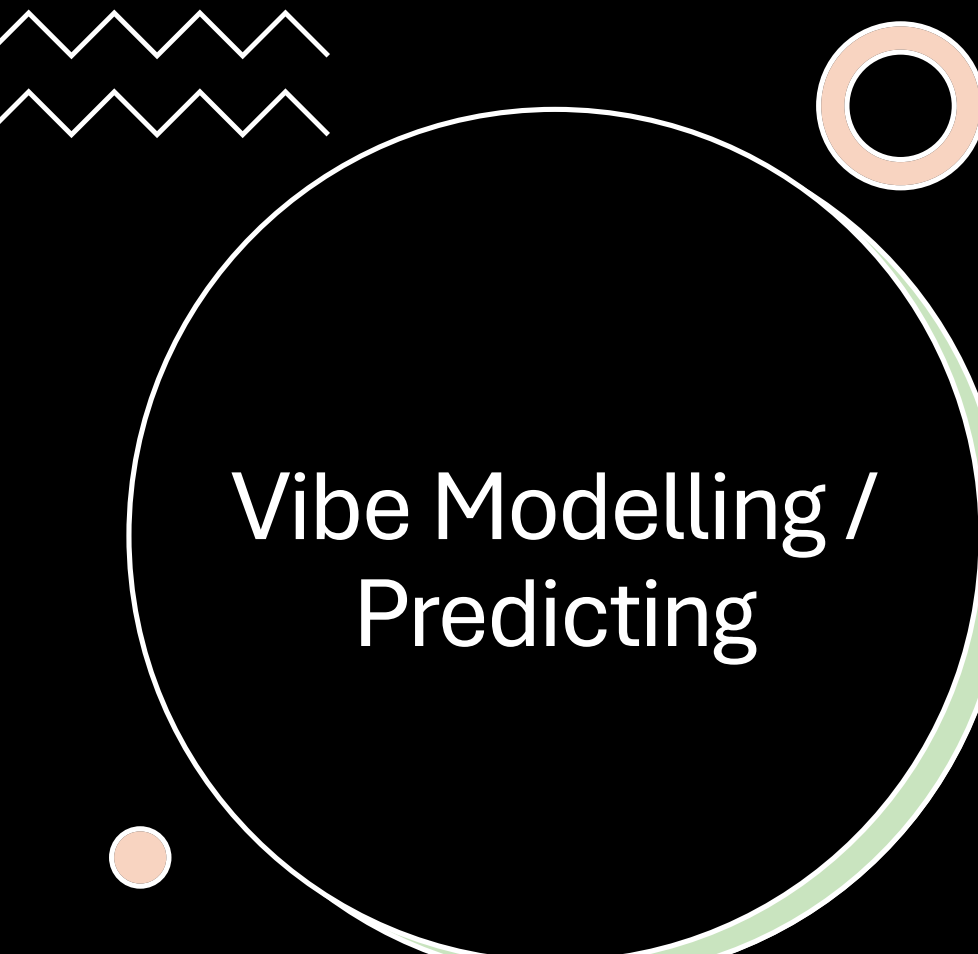




What is Vibe Coding



- Coin termed by Andre Karpathy (<https://x.com/karpathy/status/1886192184808149383>):
- *“There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. “*
- More recently it could have negative connotations where a non-developer uses AI to build software, sometimes leading to security issues or random bugs

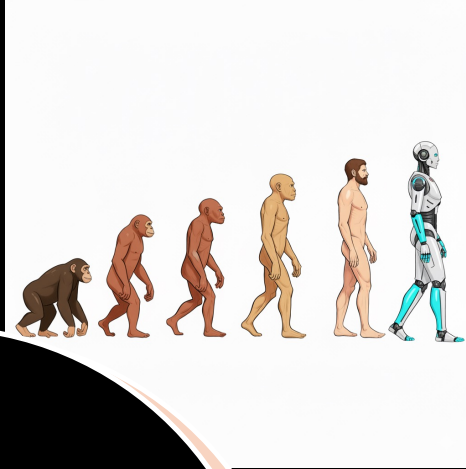


Vibe Modelling / Predicting

- Concept made up by me!
- But what if we could do the same but for Datascience?
- Can we just have AI solve for a given dataset?



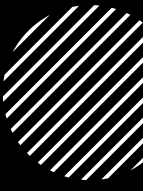
Evolution of Coding



- Looking up in manual
- Man Pages
- Ask co-worker/friend
- Open source examples
- Stack overflow
- Auto complete
- Ask AI for answer to specific problem
- Full Agentic / Vibe coding



Vibe coding tools (frontier or open weights?)



Claude Code (npm i -g
@anthropic-ai/claude-code)



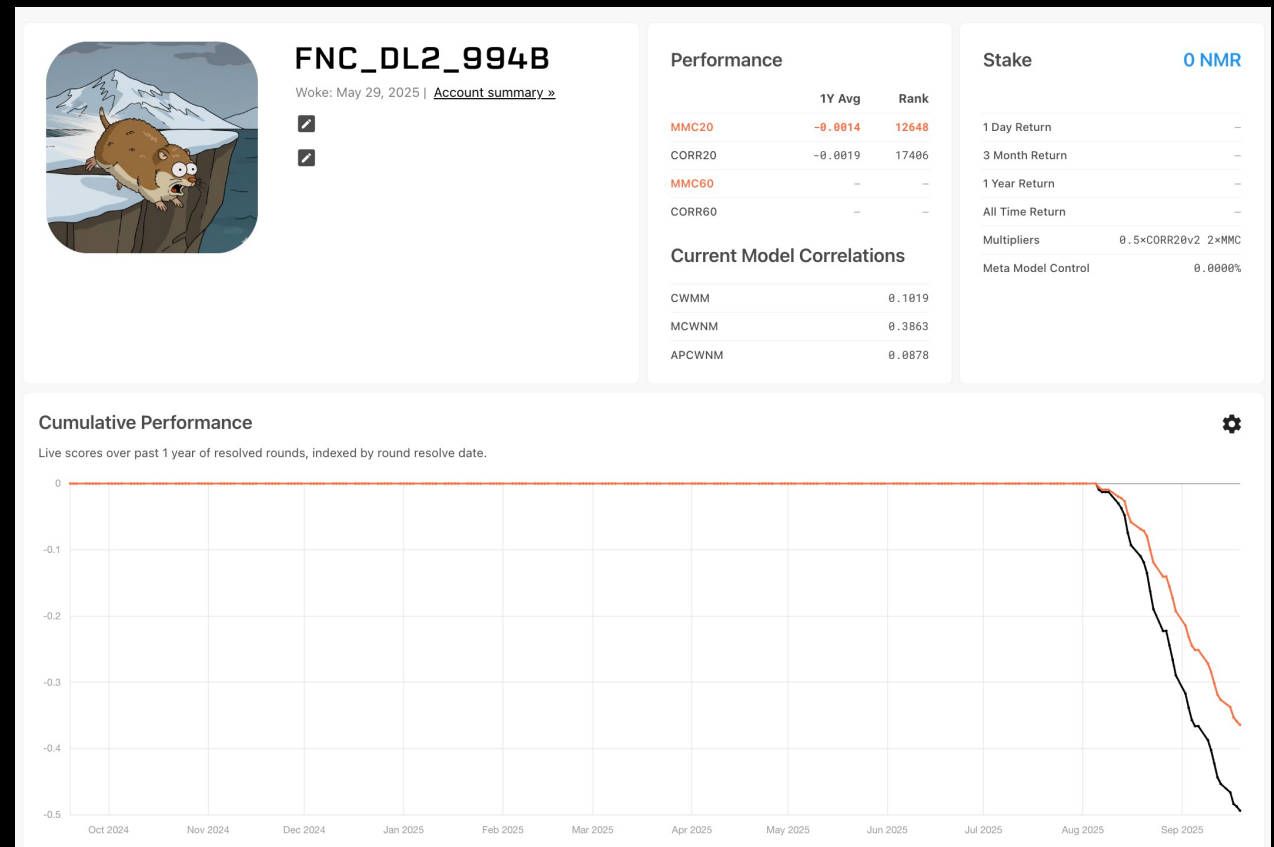
Codex (npm i -g
@openai/codex)



Gemini (npm i -g
@google/gemini)

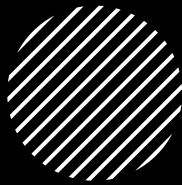
Enter Numerai – how not to do it!

- Prompt: *build me a deep learning AI model to predict values for the numerai datascience tournament.*

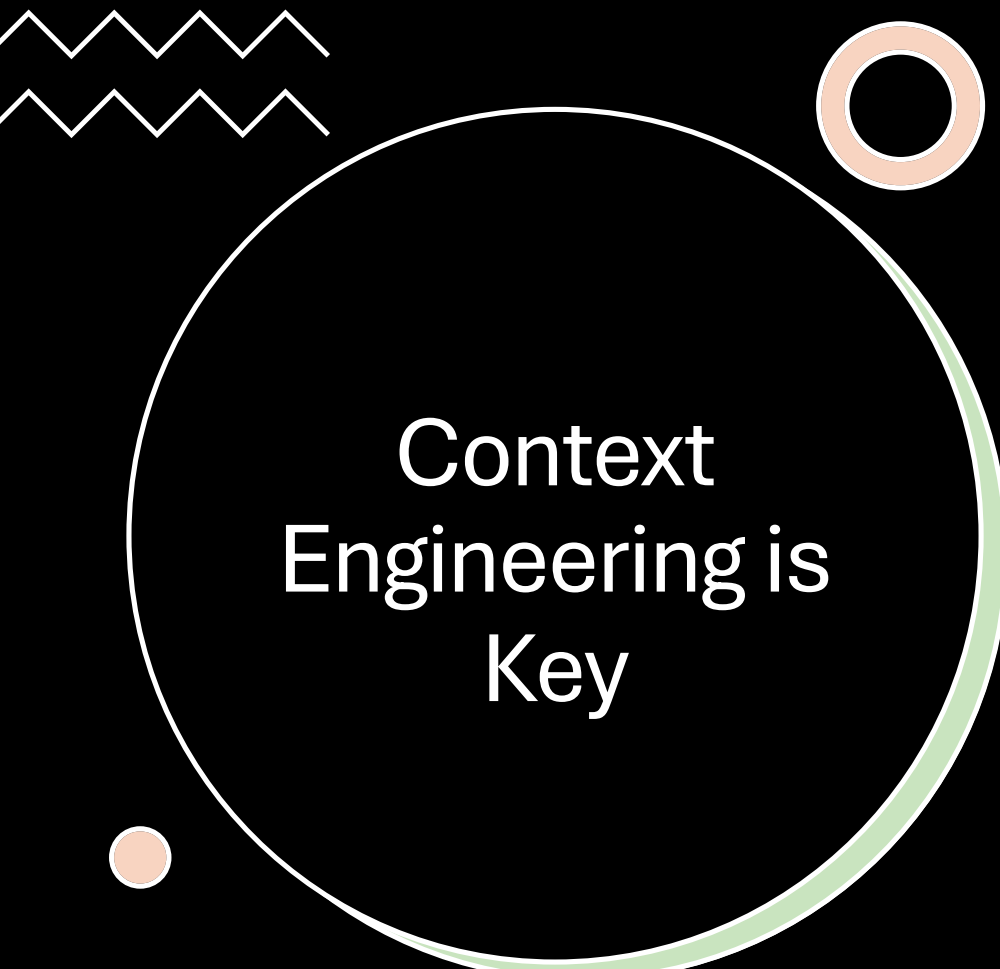




Let's work like data scientist



- Perform EDA – get some insights about the data
- Feature Engineering
- Implement models - under or over fitting – use wandb
- Fine Tune Models
- Typically done manually via Jupiter notebooks
- Just ask AI to generate python code and output EDA report
- <https://github.com/lingster/numeraire>



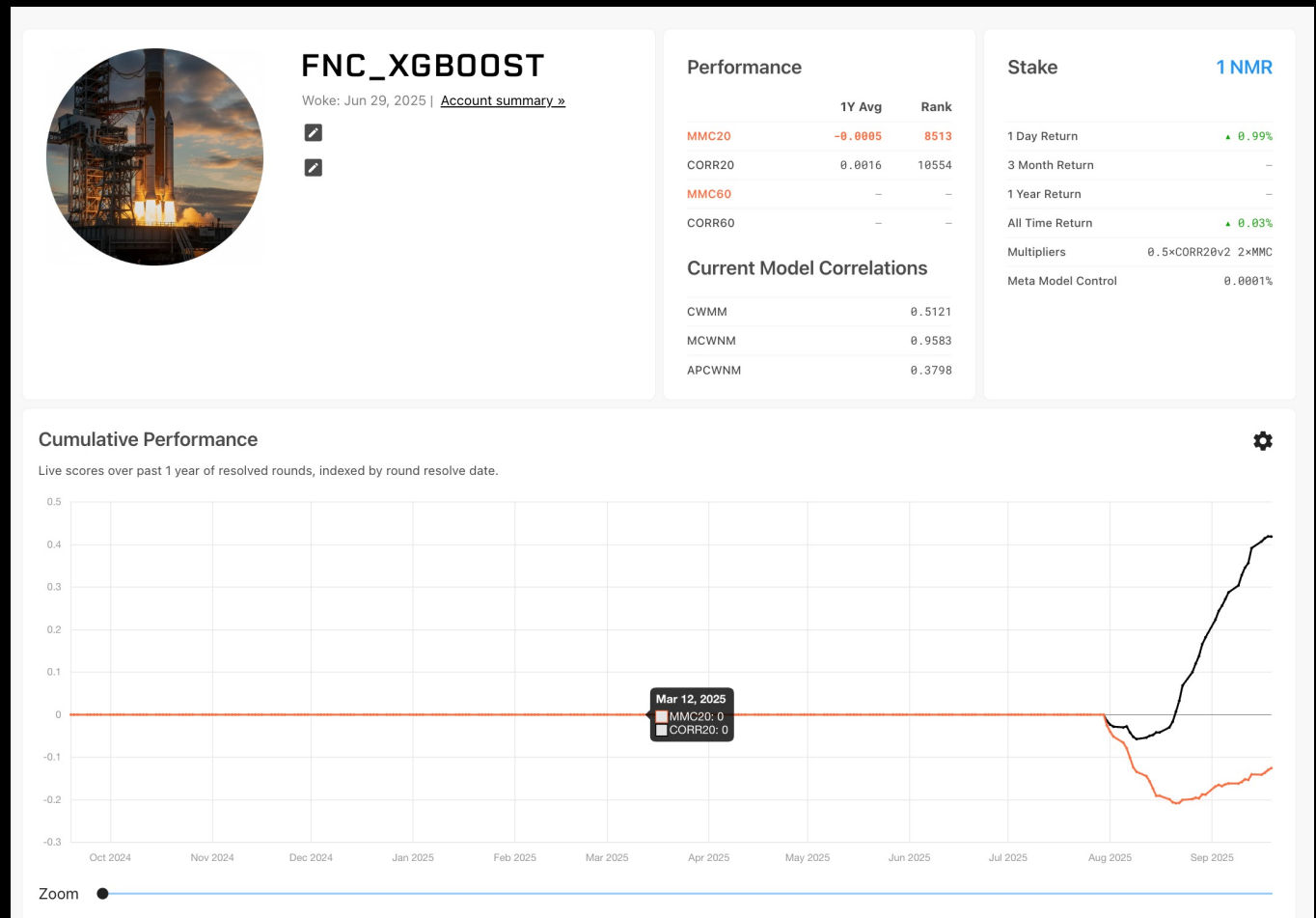
Context Engineering is Key

- Give as much background details to the LLM
- Try to give it enough instructions to focus on hone in on the result
- Try different models
- Plan with one model then execute with another



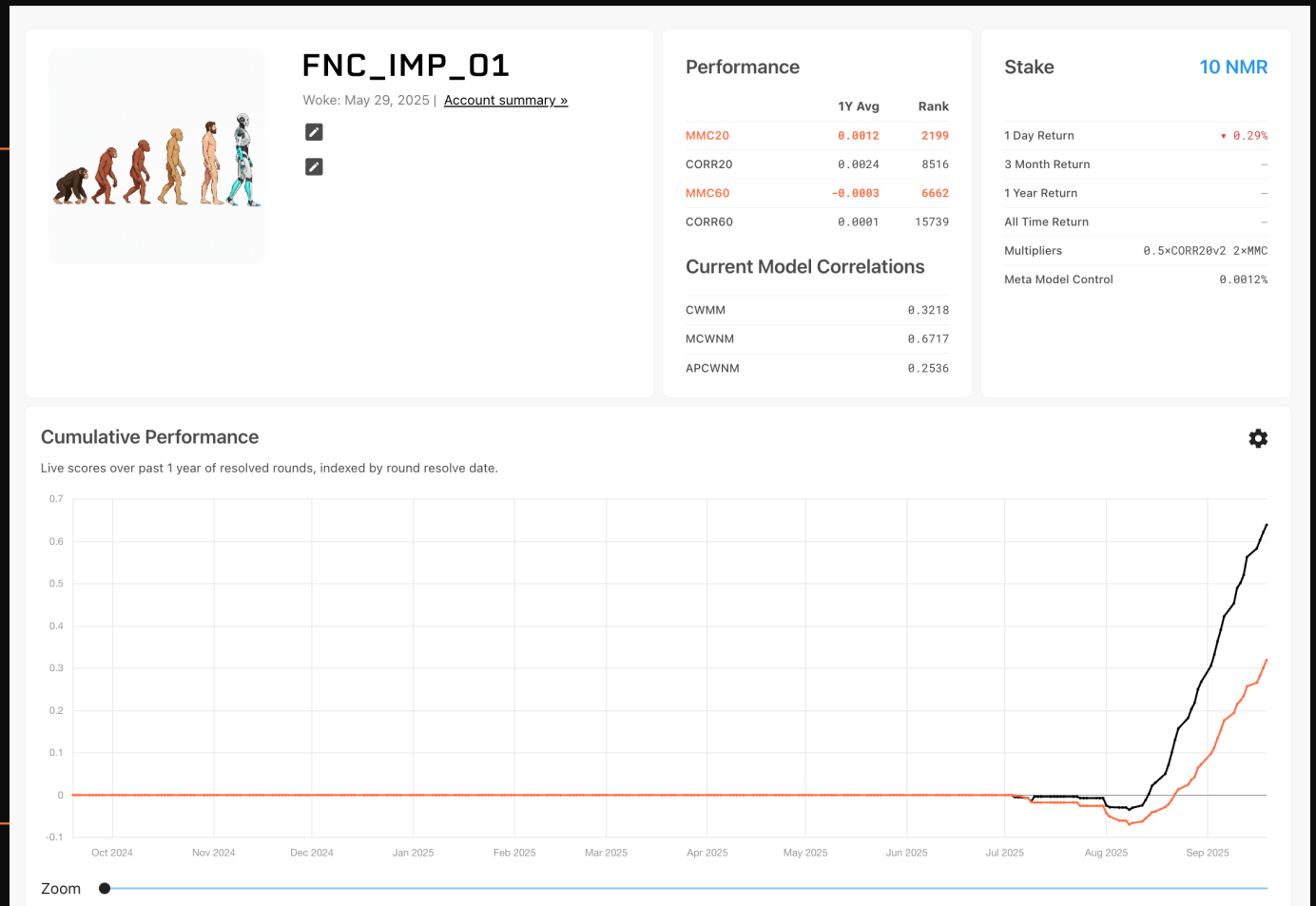
XGBoost

- Popular among kagglers
- Easy goto hence -ve MMC



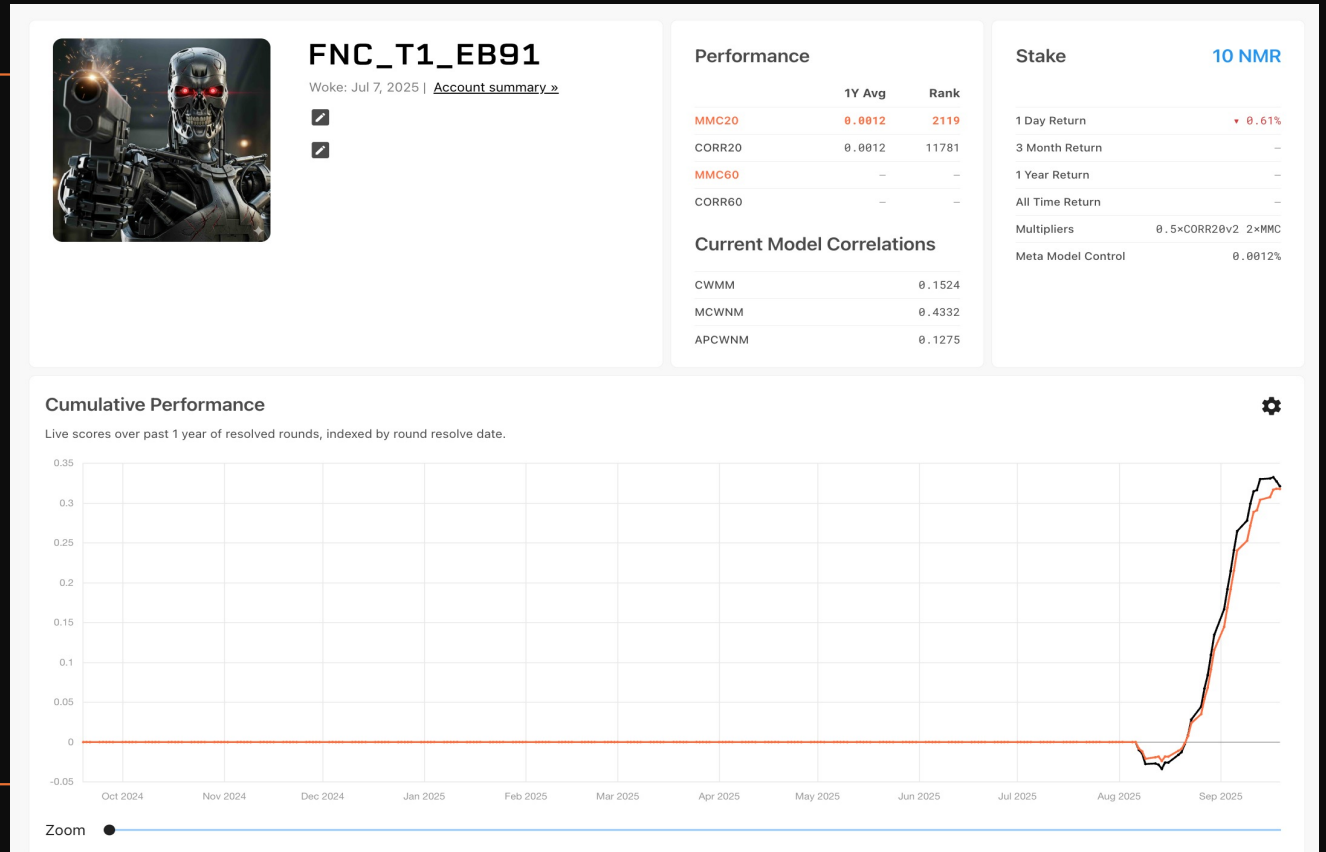
Feature Engineering Bells and Whistles

- Feature Engineering
- Ensemble of Models
- Feature Neutralization
- Hyperparameter tuning



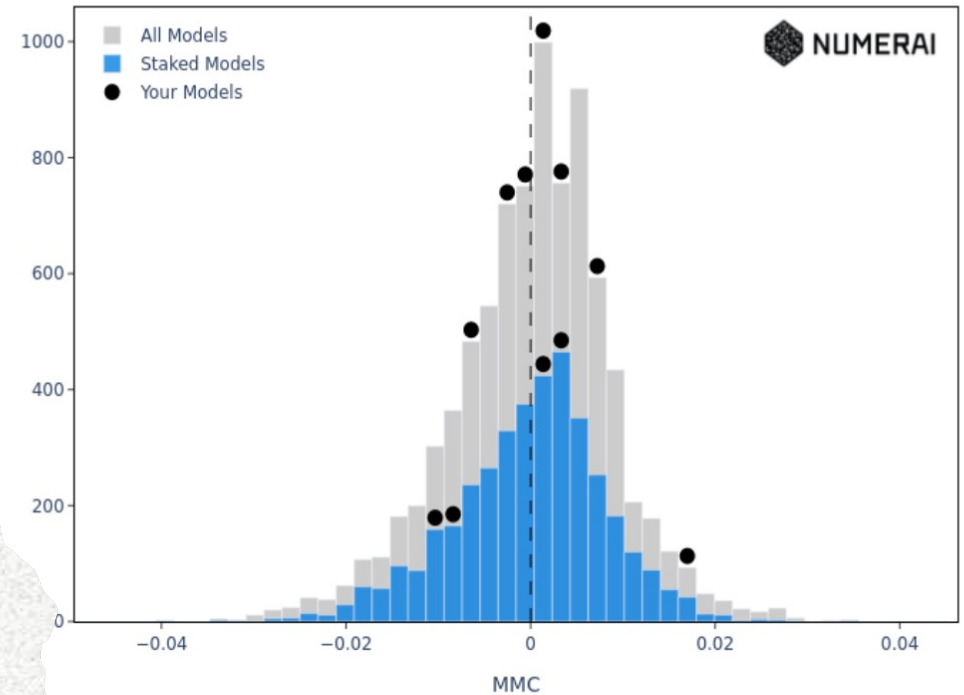
Another All singing and Dancing Model

- Build Faster
- Build More
- Fail Faster
- Experiment More
- Brute Force



Hitting the spot!

-0.0077 -0.0069 -0.0217 NMR



Model	CORR20v2	MMC	At Risk	Payout
<u>C DL C1D8</u>	-0.0083	-0.0082	9.97 NMR	-0.02 NMR
<u>C ME E D676</u>	-0.0128	-0.0098	1.0 NMR	-0.0 NMR
<u>C XGBOOST</u>	-0.0003	0.0046	0.97 NMR	0.0009 NMR
<u>C EFFICIENT</u>	0.0016	0.0028	0.41 NMR	0.0003 NMR
<u>NC IMP 01</u>	0.0171	0.0179	—	—
<u>IC NAI FN</u>	0.0059	0.0076	—	—
<u>NC IMP E</u>	0.0011	0.0048	—	—

Still more room for improvements

- Retraining of models as validation dataset grows
- Pruning – Building larger Neural networks, then pruning
- Train strategies – rolling window, experiment with sizes
- Mixture of Experts
- Ensemble/Stack of Models



Conclusions

- AI Assisted Coding helps us:
 - Build and deploy more models
 - Iterate faster
- Next steps:
 - Signals!
 - AI to extract the data
 - Then run similar process as before
 - Let's go!
 - **Grandmaster!!** 😊